

Physics You Can Feel: A Pseudo-Haptic VR Sandbox for Experiencing Mechanics and Material Properties

CS567_SVEN

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Abstract

Physical intuition about mass, volume, density, and friction, and about simple machines such as levers and pulleys, is hard to build from equations and 2D diagrams, and the demonstrations that would make it concrete are often impractical or unsafe in a classroom. We propose an immersive virtual-reality sandbox in which learners create, hold, weigh, and manipulate objects and simple machines. Because commodity headsets cannot render true weight, we use pseudo-haptic cues so a denser object feels heavier and a loaded lever feels harder to move. The application is pure VR with no in-scene agent; agentic AI is instead the development workflow, an AI coding agent that helps implement the Unity project. As an online, demonstration-focused project it has no human-subjects study: success is an interactive Quest 2 demonstration whose pseudo-haptic cues track the underlying physics across several scenarios.

Keywords

3D user interfaces; virtual reality; embodied learning; pseudo-haptics; physics education; agentic AI; AI-assisted development

1 Introduction and Motivation

Introductory physics rests on relationships that students can state long before they can predict with them. Density relates mass to volume, buoyancy determines whether an object floats, friction resists sliding in proportion to the normal load, and simple machines such as levers and pulleys trade force for distance. The usual failure is not arithmetic but intuition. Asked whether a kilogram of gold or a kilogram of ice is larger, which of two blocks slides farther down a ramp, or where a load balances on a lever, many learners fall back on misconceptions such as “heavier means bigger.” Textbooks and 2D simulations show the formula and the result, yet they do not let a student *feel* the difference.

Doing the demonstration in real life is harder than it sounds: the comparisons that isolate a single variable are exactly the ones that are awkward, expensive, or unsafe to stage, since matched-mass materials are not classroom stock and friction is hard to hold constant. Virtual reality can present these controlled manipulations on demand, but a handheld controller weighs the same whatever it represents, so the sensation many lessons depend on is missing. The

open question is whether a spatial interface can make physical properties and mechanical behavior *felt* well enough to matter, given that the dominant channel for weight and force is the one current hardware cannot deliver.

2 Related Work

Our sandbox sits at the intersection of four lines of research: immersive virtual reality (VR) for science learning, the difficulty of rendering weight and force on commodity hardware, the pseudo-haptic techniques that stand in for that missing sensation, and the interaction-design foundations that make direct manipulation legible. We review each in turn and close by stating what distinguishes our approach.

2.1 Immersive VR for Science Learning

The pedagogical case for immersive VR rests less on visual fidelity than on the sense of being present. Slater and Sánchez-Vives frame this through the place illusion (the feeling of being in the virtual space) and the plausibility illusion (the feeling that events are really happening), which together lead users to respond to virtual events as if they were real [6]. For a physics sandbox this matters directly. If a learner treats a virtual block and lever as real objects, then manipulating them can carry the intuitive weight that a 2D simulation cannot.

Empirical work in physics education adds a consistent caveat: immersion helps, but only when the learning experience is designed for it. A recent meta-analysis by Wu et al. across many head-mounted display studies finds that immersive VR yields a positive but moderate overall effect on learning performance, and that this effect is strongly moderated by factors such as subject domain, learner control, and how the experience is designed [10]. Tsivitanidou et al. report conceptual gains from an inquiry-based immersive VR physics activity, yet find that those gains interact with learners’ attitudinal profiles rather than following from immersion alone [7]. Georgiou et al. reach a compatible conclusion from the design side, showing that outcomes depend heavily on how interaction and scaffolding are structured within the experience [2]. One common way to supply that scaffolding is an embedded pedagogical agent. Makransky et al. find that guidance from such an agent can aid immersive VR science learning, but that its benefit is conditional and varies with how well the agent is matched to the learner [5]. Taken

together, these studies establish that immersive physics learning is promising but design-sensitive. The value comes from what the learner is asked to do and how the task is scaffolded, not from the headset itself. We take a deliberately different scaffolding stance. Rather than an in-scene agent, we scaffold through scene and task design, using tightly scoped scenarios that isolate a single variable so that the cause of each outcome is unambiguous.

2.2 The Missing Channel: Rendering Weight and Force

The central obstacle for a physics-of-materials experience is that the very quantity the lessons depend on, namely weight and the force needed to move a loaded mechanism, is the one commodity that VR cannot deliver. A handheld controller weighs the same whatever it represents. Wang et al. survey the state of haptic display for VR and make clear that genuine, general-purpose force feedback remains far from consumer reach, constrained by actuation, workspace, and grounding requirements that headset-and-controller setups do not meet [9]. Their survey frames the design problem for us precisely. If we want a denser object to feel heavier on a Quest 2, we cannot wait for the hardware to render the force, and must instead substitute perception for physics.

2.3 Pseudo-Haptics: Substituting Perception for Physics

Pseudo-haptics addresses this gap by exploiting the dominance of vision over proprioception. Manipulating the visual consequences of a motion, such as the control-to-display gain or a small lag between hand and object, induces a felt sense of weight, stiffness, or resistance without any force actuator. The foundational demonstration comes from Lécuyer et al., who showed that altering the visual response of an isometric input device makes users perceive force and stiffness that no actuator produces, establishing that a control-to-display manipulation alone can stand in for force feedback [4]. Ujitoko and Ban survey the design space and applications of pseudo-haptics, cataloguing how such visual manipulations reliably produce haptic sensations and how they have been applied [8]. Their taxonomy is the methodological backbone for how we map density and mechanical load onto gain and visual lag. Kim et al. extend the picture to the multisensory case, showing that added auditory and visual cues measurably shift the perceived weight of a virtual object [3], which suggests our weight illusion can be reinforced by coordinated feedback rather than resting on visual gain alone. These techniques are not limited to lifting a single object. The same principle of modulating the visual outcome of an effort extends naturally to the resistance of operating a machine, such as a lever that grows harder to move as it is loaded.

2.4 Foundations of 3D Interaction

Whatever the illusion, the learner must first be able to reach out, grab, and place objects fluidly, or the physics never gets probed. We ground our interaction design in Bowman

et al.’s canonical treatment of 3D user interfaces, which formalizes selection, manipulation, and travel as the core tasks of spatial interaction and catalogues the techniques for each [1]. This foundation informs the grab-and-place model at the heart of the sandbox and keeps our contribution, the pseudo-haptic property mapping, layered on top of well-understood manipulation techniques rather than reinventing them.

2.5 What Is Different About Our Approach

Two gaps emerge from this literature. First, immersive physics studies consistently deliver visual and interactive experiences but rarely give learners genuine haptic contrast between objects. A student may see that two blocks differ, but does not feel one resist more than the other. Second, pseudo-haptic studies convincingly render the felt weight of an isolated object, but seldom span both material properties (density, buoyancy, friction) and simple machines (levers, pulleys) within one coherent environment and interaction model. Our sandbox targets this intersection. We use pseudo-haptic gain and visual-lag modulation as a unifying mechanism that makes an unrenderable physical property perceptible across a family of controlled, single-variable scenarios, and we extend the same mechanism from holding a heavier object to feeling a loaded lever push back. Because our contribution is an interaction technique for making a physical property felt, rather than a study of learning gains, we evaluate it at the system level: whether the pseudo-haptic cues behave monotonically with the underlying physics, and whether each scenario matches its closed-form prediction.

3 Approach

We will build a VR “physics sandbox” in which the learner constructs and probes scenarios and the interface makes the governing property perceptible. Early scenarios target material properties: pick up two same-mass blocks of different materials, weigh them on a balance, drop them in water to see buoyancy, and drag them across a ramp to compare friction. The same interaction model extends to mechanical devices, such as placing weights on a lever to find its balance point or routing a rope through a pulley to feel the reduced effort. In each case a single variable is isolated so the outcome has a clear cause. Weight and resistance are conveyed through pseudo-haptics, modulating control-to-display gain and visual lag so a denser block feels heavier and a loaded lever feels harder to move [3, 8]. This is where the project complies with HCI research in **Virtual Reality**: the contribution is an interaction technique for making an unrenderable physical property perceptible through direct, embodied manipulation.

The second aspect is the **agentic-AI** development workflow, and we are deliberate about the distinction the assignment draws. The finished application has no in-scene chatbot for the learner to talk to. Instead, agentic AI is how the software is built: an AI coding agent works in the project with real tools, editing Unity C# scripts, running builds, reading

compiler and test output, and iterating toward a working scene over many steps rather than emitting one block of code from a single prompt. It plans, uses tools, observes results, and self-corrects toward a goal. We report what it implemented, where it needed human correction, and how the workflow shaped the build. The professor has confirmed that a pure VR project built this way satisfies the assignment.

4 System Vision and Technology

When it works, the demo is a first-person VR scene with a ground plane, a work area, and a menu for spawning materials and mechanical devices (Figures 1 and 2). The learner grabs objects with hand-tracking or controllers and feels a denser one resist motion or a loaded lever push back; a side panel shows each object's properties and the measured outcome of an interaction. Early scenarios cover density, buoyancy, and friction, with levers, pulleys, and projectiles added on the same framework as the project grows.

We will build in **Unity 6** with the **XR Interaction Toolkit** and **OpenXR**, in **C#**, targeting the **Meta Quest 2** (hand-tracking and controllers). Physics uses Unity's rigid-body engine with per-material density and friction parameters and standard joints for levers and pulleys. Development uses an agentic-AI coding workflow, an AI agent that edits scripts, runs builds, and reads compiler output while iterating toward each scene. We already have a Quest 2 and a Unity development machine.

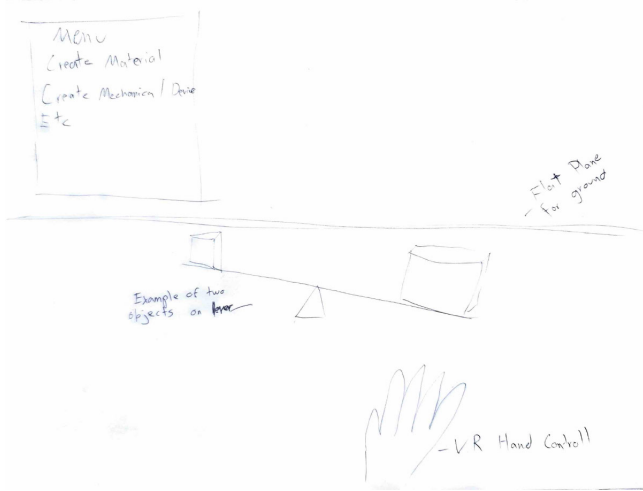


Figure 1: Hand-drawn concept sketch. From the learner's first-person view: a menu panel (create a material, create a mechanical device, and so on), two objects balanced on a lever over a fulcrum on a flat ground plane, and a tracked VR hand for direct grab-and-place interaction.

5 Evaluation Plan

This is an online, demonstration-focused project, so it has no human participants and requires no IRB submission. We



Figure 2: AI-enhanced render of the same scene (generated with Google Gemini from the sketch in Figure 1). A labeled steel block and concrete cylinder rest on a lever, showing how the learner reads object properties and manipulates them by hand while the floating menu spawns new materials and mechanisms.

evaluate the system itself, with success defined at the system level rather than by learning gains on students. **Physics fidelity:** automated checks compare each scenario against its closed-form prediction within tolerance, for example that equal-mass blocks show the correct relative volumes, objects float or sink by density, ramp sliding follows the friction coefficient, and a lever balances at the computed load position. **Pseudo-haptic behavior:** the cues must be monotonic, a denser object always feeling heavier and a more heavily loaded lever always harder to move, with gain and lag mapping to density and load in the intended direction. **Development workflow:** because agentic AI is our build method, we report which parts of the implementation the coding agent produced through tool use and iteration and where human correction was needed. **Success** is an interactive Quest 2 demonstration in which the fidelity checks pass, the pseudo-haptic cues behave monotonically across scenarios, and the scene runs stably on-device. An informal peer walkthrough may inform design, but no formal user data is collected.

6 Deliverables

By mid-November:

- A Unity 6 scene on Quest 2 with grab-and-place interaction for at least four materials, a working balance, a water tank, and a ramp, plus one mechanical device (a lever), all with pseudo-haptic weight and force modulation (gain and visual lag).
- Automated physics-fidelity checks comparing the simulation against closed-form predictions for density, buoyancy, friction, and the lever.
- In-scene logging of interactions and outcomes to file.

Toward the **early-December** presentation:

- An expanded sandbox adding at least one more concept (for example a pulley or projectile launcher) built on the same interaction and physics framework.
- A recorded demonstration video walking through each concept on-device.
- The paper extending this proposal, documenting the design, the system-level evaluation, and a report on the agentic-AI development workflow used to build it.

Roles: This is a solo project (group CS567_SVEN). Sven Skoglund is responsible for all work, including the Unity interaction and pseudo-haptics, the physics-fidelity checks, the demonstration, and directing the agentic-AI development workflow.

7 Conclusion

We are building a VR physics sandbox that turns abstract relationships, from density and friction to the mechanics of levers and pulleys, into things a learner can feel and test through direct, embodied manipulation. The application is pure VR with no in-scene agent; agentic AI is the workflow used to build it. The class should expect a working Quest 2 demonstration, automated checks that the simulation matches the underlying physics, and a report on how the agentic-AI development process shaped the result.

Use of AI

Generative AI (a large language model) was used to help locate candidate peer-reviewed references and to critique the argument and grammar of drafts. Google Gemini generated the enhanced mockup in Figure 2 from the author’s hand-drawn sketch (Figure 1). The implementation is planned as an agentic-AI development effort in which an AI coding agent assists in writing and iterating on the Unity project under the author’s direction, and this will be documented in the final paper. All prose in this proposal was reviewed and revised by the author, and every cited work was read and verified by the author.

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